

Then all investors hold the same MVE portfolio—just in different quantities depending on their own risk aversion. Since everyone holds the same MVE and this is the best portfolio that can be held by all investors, the MVE portfolio becomes the market factor in equilibrium.

Equilibrium

The equilibrium concept is extremely important. Equilibrium occurs when investor demand for assets is exactly equal to supply. The market is the factor in equilibrium because in CAPM land, everyone holds the MVE portfolio (except for those who are infinitely risk averse). If everyone's optimal risky portfolio (which is the MVE) assigns zero weight to a certain asset, say AA stock, then this cannot be an equilibrium. Someone must hold AA so that supply equals demand. If no one wants to hold AA, then AA must be overpriced and the expected return of AA is too low. The price of AA falls. The expected payoff of AA stays constant under CAPM assumptions, so that as the price of AA falls, the expected return of AA increases. AA's price falls until investors want to hold exactly the number of AA shares outstanding. Then, the expected return is such that supply is equal to demand in equilibrium. Since all investors hold the MVE portfolio, the MVE portfolio becomes the market portfolio, and the market consists of each asset in terms of market capitalization weights.

Equilibrium ensures that the factor—the market portfolio—will have a risk premium and that this risk premium will not disappear. The market factor is *systematic* and affects all assets. The market risk premium is a function of the underlying investors' risk aversions and utilities. That is, the risk premium of the market factor reflects the full setup of all people in the economy. The factors that we introduce later—tradeable factors like value-growth investing and volatility investing or macro factors like inflation and economic growth—will also carry risk premiums based on investor characteristics, the asset universe, and the production capabilities of the economy. They will disappear only if the economy totally changes. Equilibrium factor risk premiums will not disappear because clever hedge funds go and trade them—these types of investment strategies are not factors. Investors cannot arbitrage away the market factor and all other systematic factors.

3.2. CAPM LESSON 2: EACH INVESTOR HAS HIS OWN OPTIMAL EXPOSURE OF FACTOR RISK

In Figure 6.3, all investors will hold the market portfolio, just in different proportions. Pictorially, they have different proportions of the risk-free asset and the market portfolio and lie on different positions on the CAL line (see chapters 2 and 3). Thus, each individual investor has a different amount of factor exposure just as different individuals have different nutrient requirements.